An obstruction to one-parameter presentations of an elliptic point

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Abstract

For $K = \mathbb{Q}(\sqrt{2})$ and $E : y^2 = x^3 - (11 - 6\sqrt{2})x$, every algebraic curve over $\mathbb{Q}$ mapping to $\mathrm{Res}_{K/\mathbb{Q}} E$ has finite rational image. Consequently no finite collection of one-parameter algebraic presentations with finite fibers can represent an infinite subset of $E(K)$. Two rational parameters suffice, with one finite quartic choice and finitely many charts. The obstruction is unconditional and concerns presentations of the elliptic point itself; it is not a lower bound for arbitrary Diophantine definitions of the integers.

1 The elliptic curve and its conjugate

Put

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \qquad E : \; y^2 = x^3 - Dx.$$

Let σ be the nontrivial automorphism of $K/\mathbb{Q}$, and write $E^\sigma : y^2 = x^3 - \sigma(D)x$.

Proposition 1.1. *The elliptic curves E and E^σ are not isogenous over K. More precisely,*

$$\mathrm{Hom}_{K(i)}(E, E^\sigma) = 0.$$

Proof. The field $K(i)$ has a place with residue field $\mathbb{F}_{17}$ at which $\sqrt{2}$ reduces to 6 and i reduces to 4. At this place both curves have good reduction, and their equations are

$$y^2 = x^3 - 9x, \qquad y^2 = x^3 - 13x.$$

The discriminants are nonzero because they are $64 \cdot 9^3$ and $64 \cdot 13^3$ modulo seventeen. The square residues modulo seventeen are $0, 1, 2, 4, 8, 9, 13, 15, 16$. Enumeration of the seventeen abscissas gives

D mod 17	$x^3 - Dx = 0$	nonzero square	nonsquare	$\#E_D(\mathbb{F}_{17})$
9	3	8	6	20
13	3	6	8	16.

The last column includes the point at infinity. A nonzero homomorphism between elliptic curves is an isogeny. An isogeny at a place of good reduction gives an isogeny of the reductions, whose Frobenius polynomials, and therefore their numbers of rational points, are equal. The unequal counts exclude such a homomorphism over $K(i)$, hence also over K. The accompanying script reproduces the counts by two independent finite enumerations. □

The conjugate curves are nevertheless quadratic twists of one another. Indeed $d\sigma(d) = 7$, and

$$\delta = \frac{\sigma(d)}{d} = \frac{7}{d^2}$$

represents the squareclass of seven. Over $K(\sqrt{7})$ the map $(x, y) \mapsto (\delta x, \delta\sqrt{\delta}\, y)$ is an isomorphism $E \to E^\sigma$. Thus the nonisogeny assertion concerns the stated fields; it is not a claim of geometric nonisogeny.

2 Simplicity of the Weil restriction

The Weil restriction $A = \mathrm{Res}_{K/\mathbb{Q}} E$ is the abelian surface characterized by

$$A(R) = E(R \otimes_{\mathbb{Q}} K)$$

for every $\mathbb{Q}$-algebra R. In particular $A(\mathbb{Q}) = E(K)$, and $A_K \simeq E \times E^\sigma$.

Proposition 2.1. *The abelian surface A is simple over $\mathbb{Q}$.*

Proof. Suppose $B \subset A$ were a nonzero proper abelian subvariety over $\mathbb{Q}$. It would have dimension one. The two projections of B_K into $E \times E^\sigma$ are homomorphisms of elliptic curves. If both were nonzero, both would be isogenies, making E and E^σ isogenous over K, contrary to Proposition 1.1. Exactly one projection is therefore nonzero. The other is zero, so B_K is one of the two factors. But the nontrivial Galois automorphism interchanges those factors. Neither factor descends to a $\mathbb{Q}$-subvariety of A, a contradiction. □

Lemma 2.2. *The group $E(K)$ is infinite. In particular the point*

$$P = \left(\frac{57 - 28\sqrt{2}}{8}, \frac{-660 + 417\sqrt{2}}{32} \right)$$

has infinite order.

Proof. Its coordinates satisfy the curve equation. The prime three is unramified in K, with residue field $\mathbb{F}_9$, and the curve has good reduction there. Normalize its valuation by $v(3) = 1$. The displayed point has $v(x(P)) = 0$ and $v(y(P)) = 1$, while $3x(P)^2 - D$ is a unit. The tangent slope therefore has valuation -1, and the duplication formulas give

$$v(x(2P)) = -2, \qquad v(y(2P)) = -3, \qquad v(z(2P)) = 1, \qquad z = -x/y.$$

The odd integral formal multiplication series $[m](Z) = mZ + Z^3 g_m(Z^2)$ now shows that multiplication by an integer prime to three preserves a positive parameter valuation, while multiplication by three raises it by one. The higher terms have strictly larger valuation than the linear term. Consequently $v(z([2n]P)) = 1 + v_3(n)$ for $n \neq 0$, and no nonzero multiple of $2P$ is the identity. Therefore P has infinite order. □

3 Images of algebraic curves

Theorem 3.1. *Let F be a number field and let B/F be an abelian variety which has no elliptic abelian subvariety defined over F. If V is an algebraic set over F of dimension at most one, $U \subseteq V$ is open, and $f : U \to B$ is a morphism, then $f(U(F))$ is finite.*

Proof. There are finitely many irreducible components. A zero-dimensional component contributes finitely many points. If an irreducible curve is not geometrically integral, all its F-points lie in the finite intersections of its distinct conjugate geometric components. It therefore suffices to consider a geometrically integral curve.

Take its smooth projective normalization C. Outside finitely many points the normalization is an isomorphism, so rational points lift. The induced rational map $C \dashrightarrow B$ extends to a morphism on C because B is proper. If C has genus at least two, Faltings's theorem makes $C(F)$ finite [2].

If C has genus zero and has an F-point, it is isomorphic to $\mathbb{P}^1_F$. Every morphism from $\mathbb{P}^1$ to an abelian variety is constant [1, Chapter I, Proposition 3.9]. If $C(F)$ is empty there is nothing to prove.

Finally suppose C has genus one and an F-point c_0. Then (C, c_0) is an elliptic curve. Subtracting $f(c_0)$ from f gives a homomorphism of abelian varieties [1, Chapter I, Corollary 1.2]. If it were nonzero, its image would be an elliptic abelian subvariety of B defined over F, contrary to the hypothesis. Hence f is constant. The finitely many points omitted in passing to the normalization only add a finite set of image values. □

Corollary 3.2. *Let $C_1, \dots, C_r$ be algebraic sets over $\mathbb{Q}$, each finite over a nonempty open subset of $\mathbb{A}^1_{\mathbb{Q}}$. For any rational maps $f_j : C_j \dashrightarrow A$, the union of the images of their rational points, wherever the maps are defined, is finite. In particular it cannot contain an infinite subset of $E(K) = A(\mathbb{Q})$.*

Proof. Every C_j has dimension at most one. Apply Theorem 3.1 and Proposition 2.1, then take the finite union. □

The corollary applies in particular to a retained rational parameter t followed by any finite list of triangular monic equations

$$F_j(t, z_1, \dots, z_j) = 0, \qquad F_j \text{ monic of positive degree in } z_j,$$

with rational functions of the resulting coordinates specifying the point of $E(K)$. Nonzero denominator conditions restrict to an open subset and do not change the conclusion. Arbitrarily high degrees and finitely many alternative charts do not avoid the finiteness obstruction.

4 A sharp two-parameter presentation

Proposition 4.1. *The set $E(K)$ admits a finite collection of presentations using two retained rational parameters and one finite algebraic root of degree at most four.*

Proof. Retain $a_0, a_1 \in \mathbb{Q}$ and put $a = a_0 + a_1\sqrt{2}$. Write $a^3 - Da = A_0 + A_1\sqrt{2}$, where

$$A_0 = a_0^3 + 6a_0a_1^2 - 11a_0 + 12a_1,$$
$$A_1 = 3a_0^2a_1 + 2a_1^3 - 11a_1 + 6a_0.$$

Let Y be a rational root of

$$Y^4 - 4A_0Y^2 + 8A_1^2 = 0, \qquad Y \neq 0.$$

Then

$$(a,\ Y/2 + (A_1/Y)\sqrt{2})$$

is a point of $E(K)$. Conversely it represents every point whose ordinate has nonzero rational component.

Put $T_0 = (0, 0)$. If $Q = (a, b)$ is not a two-torsion point, at least one of $Q, Q + T_0$ has a nonzero rational component in its ordinate. To see this, suppose both ordinates were in $\sqrt{2}\mathbb{Q}^\times$. The addition formula gives $b(Q + T_0)/b(Q) = D/a^2 \in \mathbb{Q}$. Thus $a/d \in \mathbb{Q} \cup \sqrt{2}\mathbb{Q}$. The equation $b^2 = d^3(a/d)((a/d)^2 - 1) \in \mathbb{Q}^\times$ would then imply $d^3 \in \mathbb{Q} \cup \sqrt{2}\mathbb{Q}$, whereas $d^3 = 45 - 29\sqrt{2}$ has both components nonzero. This is a contradiction.

Use the displayed quartic chart directly, or translate its output by T_0. The remaining points are the identity and the three nonzero two-torsion points, which are finite fixed alternatives. This gives the asserted presentation. □

The one-parameter obstruction and the two-parameter construction give an exact minimum for this point-presentation problem. They do not give a lower bound of 29 for a Diophantine

definition of $\mathbb{Z}$ over $\mathbb{Q}$. In particular, the theorem does not assume that several elliptic points must be encoded independently in every possible defining polynomial.

The finite certificates are reproduced by the standalone script `verify_elliptic_chart.py`, using only Python's standard library. It checks the two point counts independently, the displayed elliptic points and their valuations at three, the twist identities, and examples of the quartic chart. The general finiteness and sharpness assertions follow from the proofs above.

References

[1] J. S. Milne, *Abelian Varieties*, version 2.0, 2008, Chapters I and IV. https://www.jmilne.org/math/CourseNotes/AV.pdf.

[2] G. Faltings, Endlichkeitssätze für abelsche Varietäten über Zahlkörpern, *Invent. Math.* **73** (1983), 349–366; erratum, **75** (1984), 381. https://doi.org/10.1007/BF01388432.